

Optical Fiber Communication

Introduction

Optical fiber communication transmits information as light pulses through thin glass or plastic fibers, enabling high-speed and long-distance communication.

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Working Principle

A transmitter converts electrical signals into light, which travels through the fiber by total internal reflection and is converted back into electrical signals at the receiver.

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Components

The main components are transmitter, optical fiber cable, optical amplifier or repeater, and optical receiver.

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Types of Optical Fiber

Single-mode fiber supports long-distance high-speed communication, while multimode fiber is suitable for short-distance communication.

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Advantages

High bandwidth, low attenuation, immunity to electromagnetic interference, high security, lightweight, and long transmission distance.

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Disadvantages

Higher installation cost, fragile fibers, difficult splicing, and specialized maintenance.

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Applications

Used in internet backbone networks, FTTH, cable TV, medical imaging, military communication, industrial automation, and data centers.

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Comparison with Copper Cable

Optical fiber provides higher speed, longer distance, lower signal loss, and better security than copper cables.

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Future Scope

Optical fiber supports 5G backhaul, cloud computing, IoT, smart cities, AI data centers, and next-generation communication networks.

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Viva Questions

What is total internal reflection? Difference between single-mode and multimode fiber? List advantages and applications of optical fiber communication.

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